01. AI Engineer Intern Take-home Assignment 2.0

Role: AI Engineer Intern
Product Focus: **Building the "AI Co-Worker" Engine**

Time Expectation: ~3 Days
Stack (flexible): Python, FastAPI, LangChain/LangGraph/AutoGen (optional), FAISS/Milvus/Pinecone (choose 1), any LLM API (OpenAI / Claude / Gemini / Mistral, DeepSeek, etc.)

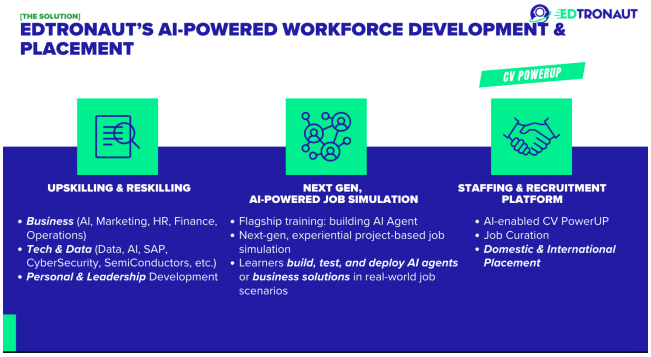
Submission Format: PDF Report / Deck + GitHub Gist/Repo
What we’re assessing:

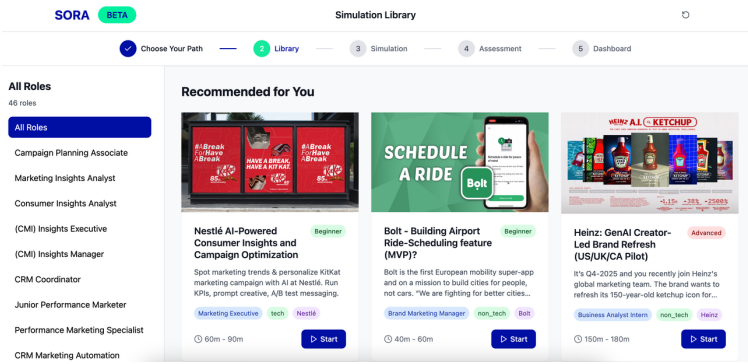
- AI Engineering and AI ops
- ***AI System Design Thinking***
- Critical Thinking, & Learning agility

Submission Instructions

Please submit your case study (PDF format, max 10–15 slides) to **hr@edtronaut.ai** and copy (cc) **tram@edtronaut.ai** within **3 days** of receiving this case.
You may use AI tools. Please note that since AI is accessible to everyone, it's the opportunity to assess candidate's critical thinking, prompt engineering skill and domain expertise, if any.

1. ABOUT EDTRONAUT





Edtronaut is redefining workforce readiness through an **AI-enabled Job Simulation Platform.** *These simulations are designed to mirror real-world job tasks, enabling learners to gain practical experience and develop job-ready skills with the key features and information as* ***(a) Hands-on Experience*** *through real-world scenarios and interaction (with co-workers, stakeholders, etc.) simulation,* ***(b) Tangible Outcomes*** *with portfolio of simulations that users can share and showcase to others, including potential employers, to demonstrate their skills and* ***(c) Social Sharing*** *that users can share externally across different social platforms like LinkedIn, Facebook, Threads, to boost their professional visibility.*

Edtronaut's website - Homepage and key products:

- Edtronaut's HP: https://edtronaut.ai/
- Edtronaut's Job Simulation (Beta): https://job-simulations.edtronaut.ai/ed
- Edtronaut's CV PowerUP: https://edtronaut.ai/cv-powerup

2. THE ASSIGNMENT (THE ASK)

2.1 The Mission

We’ ve built our **AI Assessment Engine** to analyze learners’ submissions and provide rubric-based scoring and narrative feedback (overall score, criteria breakdown, “What you did well / What to improve”).

Our next step: enable Users / Simulation Takers to **work, talk, and collaborate with AI Co-workers and Stakeholders** inside the simulation, just like how people collaborate with colleagues, cross-function teams to solve real business problems in real world.

Think of these AI entities as **NPCs (Non Player Characters) in a game**, but for the workplace, for example: A demanding marketing director, A skeptical data scientist, A friendly but busy project manager

Your Challenge: **design and prototype the AI engine** behind these co-workers.

Design the architecture for an **AI Co-worker Engine** (like ***AI Non-Player Character (NPC) Engine in Gaming***) that powers these virtual colleagues. We want you to ***conceptualize how we can create AI agents*** that don't just "answer questions" but ***act with personality, memory, and specific business functions and goals to collaborate*** with the user.

2.2 The Scenario Context

The team wants to pilot a new simulation track where learners practice different skill sets by chatting with AI co-workers.

Here is the Simulation Example for you to design the **AI Co-worker Engine** so that Simulation Taker can work and interact with. This is the Simulation Content of HRM Talent & Leadership Development - Gucci Global Group. ***Please note that this is a specific example for your ease of comprehension. In reality, the AI Engineer will need to build an AI Co-worker Engine that would work for cross simulations, not just one single simulation like this.***

08. HRM Talent & Leadership Development - Gucci 2.0

🎯 AI Environment - AI Co-worker

AI coworkers (toggle on/off):

1. **AI Co-worker 1 - Gucci Group CEO**: hold insights about Gucci Group, its mission, company culture and NDA and can defend the Group DNA, direction for each brand autonomy vs. group needs.
2. **AI Co-worker 2 - Gucci Group CHRO:** provide direction about Group HR's mission
   - (a) identify and develop talent and
   - (b) increase inter-brand mobility while
   - (c) supporting (not imposing on) brand DNA.) and
   - the Competency Framework (Vision, Entrepreneurship, Passion, Trust)
3. **AI Co-worker 3 - Employer Branding & Internal Communications Regional Manager:** who can share regional insights about the current status, specific status from the brand company regarding competency framework and potential training needs and challenges for rolling out the new competency and process.

AI tools in-sim: prompt library (headlines, disclaimers), KPI calculator, simple A/B simulator, and a one-click "portfolio pack" export (plan + posts + exec update) — employer-readable outputs per your principle.

Safety guardrails: AI suggestions are drafts; learners must confirm sources; no wagering language; use neutral phrasing. (Consistent with Microsoft's public comms

around responsible features.)

2.3 [ASK] The Deliverables

Please provide a solution covering the following four sections. We value system thinking, clarity, and creativity over perfect code.

Part 1: Persona & Interaction Design of the AI Co-worker(s) / NPC (The Logic)

How do we make AI characters feel like real co-workers (no video generation required)?

- **Persona Definition**: Define the system prompt/profile for the chosen Co-worker/NPC.

Example: What are their hidden constraints? (e.g., "The Engineering Lead is helpful but refuses to work weekends and hates changing APIs").

- **Dialogue Flow:** Provide a scripted example of a "Good" interaction vs. a "Bad" interaction between the User (Simulation Taker) and the Co-worker/NPC.
- **State Management**: How does the Co-worker/NPC remember the conversation?

Example: If the User / Simulation Taker annoys them in Turn 1, how does that affect Turn 5?

Part 2: System Architecture (The Engine)

How do we build this at scale (model to make it replicable across different simulations) ?

- **High-Level Diagram**: Sketch the architecture (Boxes & Arrows are acceptable). Show how the User Front-End talks to the Orchestration Layer and the LLM.
- **Tool Use**: Does the Co-worker / NPC need access to tools? (e.g., Can the "Engineering Lead" Co-worker / NPC actually look up a fake JIRA ticket? How would you implement that?)
- **Latency vs. Quality**: Real-time chat needs speed. How would you handle the trade-off between a complex RAG (Retrieval-Augmented Generation) pipeline and a fast response time?

Part 3: The "Director" Layer - Supervisor Agent (Orchestration)

In a game, a 'Director' controls the pacing. In our simulation, an AI must ensure the User / Simulation Taker stays on track.

- Propose a method for a Supervisor Agent (invisible to the user) that monitors the chat.
- **Scenario**: The User / Simulation Taker is getting stuck and the conversation is going in circles. How does your system detect this and trigger a subtle hint from the Co-worker / NPC to keep the simulation on track?

Part 4: Prototype / Implementation Strategy

Show us you can build it. ***In case of time constraint, you can focus on 1 AI Co-worker out of the 3 from Gucci Simulation. You have the flexibility to choose any of the 3.***

- **Tech Stack:** Which specific frameworks would you use? (e.g., LangChain, AutoGen, OpenAI Assistants API, local Llama-3, etc.) and why?
- **Pseudocode or Snippet**: Write a small Python function or pseudocode class (e.g., `class NPCAgent`, a runnable script that takes `(persona_id, user_message)` and returns `(assistant_message, state_update, safety_flags)`) demonstrating how you would initialize the character with its persona and handle the User / Simulatio Taker's message.

2.4 Evaluation Criteria

We are not looking for a fully functional product / engine, but a **feasible engineering plan**. We will assess you on:

1. **Role-Playing Fidelity**: Does your design allow for distinct, consistent AI personalities and business functions/goal that character is supposed to be, or does it just sound like ChatGPT?
2. **Architecture Soundness**: Is your technical approach systematic, scalable and modern?
3. **Problem Solving**: Did you anticipate edge cases? (e.g., What if the User / Simulation Taker tries to "jailbreak" the AI Co-worker / NPC or talks about unrelated topics?, etc.)

3. APPENDIX

3.1 FAQs

- **Can I use any vector DB or reranker?** Yes—justify your choice and show the impact (even small, e.g., hits@k).
- **What if I can’t meet the latency/cost targets?** Tell us where the bottleneck is, show metrics, and propose mitigations.
- **What if I don’t have paid API credits?** You may mock token usage/cost and explain what would change with a paid tier.

3.2 Competitive Landscape

Reference for other Upskilling, Reskilling and Workforce Development & Infrastructure

Category	Player / Example	Revenue Model	Key Strength
Career Platform & Job Matching	Handshake (US) (Students) LinkedIN (Professionals)	Freemium (B2C), Subscription (B2B)	Network effects, large-scale university reach
Job Simulations & Practical Experience	Forage (US), Reforge (US), Anthropos (US & EU)	Freemium (B2C), Corporate B2B2C	Scalable and free / fremium for learners
Internship & Placement Platform	VirtualInternships (AU & US)	Paid Programs (B2C), Partnerships (B2B)	Global reach, monetizes demand and supply
HRTech blend EdTech (Simulation)	Paradox.ai (US) Ourteam (SEA)	Corporate B2B Freemium (B2C), Corporate B2B	Network effects, large B2B base Startups clients